A CASE-STUDY BASED REPORT

ON

Wine Quality Prediction Using Decision Tree, Random Forest, and Gradient Boosting

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1. Abstract

Wine quality prediction is an important application of supervised machine learning that assists in evaluating wine samples based on measurable physicochemical properties. Because wine quality is governed by a complex mixture of chemical attributes rather than by a single isolated measurement, automated data-driven modeling offers a consistent and objective complement to traditional laboratory and sensory evaluation. This case-study project develops and benchmarks a practical prediction system around the three core tree-based supervised learning algorithms specified for the course project: **Decision Tree**, **Random Forest**, and **Gradient Boosting**.

The study utilizes the UCI Red Wine Quality dataset containing 1,599 raw records and 11 physicochemical input features: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol content. The workflow begins with comprehensive exploratory data analysis (EDA), data-type verification, missing-value inspection (0 nulls), duplicate auditing (removing 240 repeated rows to leave 1,359 unique samples), feature distribution analysis, Pearson correlation analysis, and interquartile range (IQR) outlier review. Quality scores (3 to 8) are converted into a binary classification target using the project threshold of **6.5**, segregating **Standard Quality** (quality < 6.5; 1,175 samples, 86.5%) from **Premium Quality** (quality ≥ 6.5; 184 samples, 13.5%).

Following an 80%/20% stratified train-test split (1,087 training and 272 testing records) and feature standardization via `StandardScaler`, the three model families are trained both as baselines and tuned estimators using stratified 3-fold cross-validation and `RandomizedSearchCV` optimizing for the F1-score. An additional **Weighted 3-Model Consensus** layer combines the predicted Premium-class probabilities of the three tuned classifiers using weights proportional to their cross-validation F1 performance (0.327 for Decision Tree, 0.350 for Random Forest, and 0.323 for Gradient Boosting) and an out-of-fold selected decision threshold of 0.35. On the held-out 20% test set, the tuned Gradient Boosting model achieved the highest overall accuracy (**88.97%**), precision (**0.667**), and ROC-AUC (**0.893**), whereas the tuned Decision Tree achieved the highest minority-class recall (**0.811**) and balanced accuracy (**0.791**). The Weighted 3-Model Consensus achieved the highest F1-score (**0.532**) and PR-AUC (**0.601**), with recall of **0.568**. Native tree explainability confirms that alcohol content and sulphates are the primary chemical drivers across all three models. The complete system is deployed via an interactive **Streamlit** web application for real-time quality screening.

2. Introduction

Wine quality assessment is a critical process in the food and beverage industry, as the commercial value, certification tier, and sensory profile of wine are directly influenced by interdependent physicochemical properties such as tartaric and acetic acidity, alcohol content, pH, sulphates, chlorides, residual sugar, liquid density, and sulfur dioxide levels. Traditionally, wine quality is evaluated through wet-chemistry laboratory analysis paired with sensory examination by human expert panels (sommeliers and enologists). However, human sensory panels are time-consuming, costly to scale across large production volumes, and susceptible to subjective judgment and palate fatigue.

This project, *"Wine Quality Prediction Using Decision Tree, Random Forest, and Gradient Boosting,"* treats wine quality assessment as a supervised classification problem in which routine physicochemical measurements are mapped to an objective quality tier. Rather than abandoning the requested algorithms for unrelated black-box architectures, this study keeps Decision Tree, Random Forest, and Gradient Boosting at the center of the system while making the experimental comparison rigorous and leak-free. Specifically, duplicate records are purged prior to partitioning, severe class imbalance is explicitly addressed through stratified cross-validation and F1-driven hyperparameter tuning, multiple complementary performance metrics (Accuracy, Precision, Recall, F1-Score, Balanced Accuracy, ROC-AUC, and PR-AUC) are reported, and a transparent probability-weighted consensus engine is deployed within an interactive web portal.

3. Background

A thorough review of ten prior studies utilizing the UCI Wine Quality dataset was conducted to establish the technical background and identify methodological gaps addressed by our implementation:

- **Cortez et al. (2009):** The foundational wine-quality study modeled wine preferences from physicochemical properties using Multiple Regression, Support Vector Machines (SVM), and Neural Networks. However, it did not evaluate the three tree-based ensemble classifiers (Decision Tree, Random Forest, and Gradient Boosting) investigated in our project.
- **Dahal et al. (2021):** Compared Linear Regression, Ridge Regression, SVM, Random Forest, and Gradient Boosting on the UCI red wine dataset. Because their study framed the problem primarily as continuous regression, it obscured the binary certification cutoffs required in industrial quality grading.
- **Jain et al. (2023):** Evaluated Decision Tree, Random Forest, AdaBoost, Gradient Boosting, and XGBoost on the 1,599-sample red wine dataset, confirming the effectiveness of tree models, but lacked leak-free deduplication and disciplined stratified cross-validation hyperparameter tuning.
- **Ye (2023):** Compared Decision Tree, Random Forest, Naive Bayes, and KNN classifiers, but restricted evaluation primarily to raw accuracy without reporting precision, recall, or F1-scores to inspect minority-class detection.
- **Kaur et al. (2025) & Recent Ensemble Studies (2023–2025):** Surveyed literature also includes fuzzy-logic hybrids (Kaur et al., 2025), deep neural networks, and SMOTE oversampling studies. Our literature review noted that deep learning models underperformed traditional tree ensembles on this dataset size while increasing computational overhead, and synthetic oversampling (SMOTE) risks introducing artificial chemical bias. Conversely, studies by **Udorji (2024)** and **Liu (2024)** highlighted that stratified validation, class-weighting, and hyperparameter tuning in Random Forests yield superior reliability.

Table 1: Summary of Surveyed Literature and Identified Research Gaps

Author(s) & Year	Dataset Used	Models Evaluated	Observed Limitation & Our Project Solution
Cortez et al. (2009)	UCI Vinho Verde Wine Dataset	Multiple Regression, Neural Nets, SVM	No tree-based ensembles evaluated. *Solution:* Benchmark Decision Tree, Random Forest, and Gradient Boosting.
Dahal et al. (2021)	UCI Red Wine Dataset	Linear/Ridge Regression, SVM, RF, GB	Regression framing obscures binary grading. *Solution:* Formulate 6.5 binary industrial cutoff.
Jain et al. (2023)	UCI Red Wine (1,599 samples)	DT, RF, AdaBoost, GB, XGBoost	No leak-free deduplication or CV tuning. *Solution:* Purge 240 duplicates; 3-fold Stratified RandomizedSearchCV.
Ye (2023)	UCI Wine Quality Dataset	Decision Tree, Random Forest, NB, KNN	Evaluated accuracy only; minority dropped. *Solution:* Report Precision, Recall, F1, Balanced Acc, ROC-AUC, PR-AUC.
Udorji (2024) & Liu (2024)	UCI Wine Quality Datasets	Random Forest Classifier	Highlighted class-weighting and tuning. *Solution:* Apply balanced class weights and F1-weighted 3-model consensus.
Kaur et al. (2025)	UCI Wine Quality Dataset	Fuzzy Logic + ML Hybrid Models	High complexity and reduced explainability. *Solution:* Preserve compact 3-tree scope with native feature importance.

4. Motivation for the Project

The motivation for undertaking this project rests on three practical and technical pillars:

1. **Need for Objective, Fast Enological Quality Screening:** Commercial wineries routinely measure chemical properties (acidity, pH, density, sulfur dioxide, alcohol) during fermentation and bottling. Automating the mapping from these 11 chemical measurements to a quality tier provides instantaneous, fatigue-free decision support that complements human tasting panels.
2. **Resolving Data Leakage and the Accuracy Paradox Under Imbalance:** In the raw 1,599-record red wine dataset, 240 rows (15.0%) are exact duplicates. Leaving duplicates in the dataset prior to splitting causes identical records to appear in both training and test sets, artificially inflating test accuracy. Furthermore, after applying the 6.5 quality threshold, 86.5% of unique samples are Standard quality while only 13.5% are Premium quality. A classifier that reports only raw accuracy can appear strong (~86.5%) while failing to detect rare Premium vintages. This motivated our focus on duplicate purging, stratified cross-validation, and F1/Recall/PR-AUC benchmarking.

3. **Combining Complementary Tree Paradigms via Explainable Consensus:** Single Decision Trees, bagged Random Forests, and boosted Gradient Boosting trees view the same chemical data differently and make non-identical errors. Combining their probability outputs through a cross-validation F1-weighted consensus layer—and exposing feature importances inside a web UI—provides a more trustworthy system than relying on any single model alone.

5. Overview of the Work Carried Out

The complete work carried out in this project encompasses six integrated stages:

1. **Exploratory Data Analysis (`Wine_Quality_EDA.ipynb`):** Conducted a 17-step programmatic exploration of `winequality-red.csv` covering schema verification, missing-value heatmaps, duplicate counting, descriptive statistics, target score distribution, 11-feature histograms, Pearson correlation matrices, quality-stratified boxplots, pairplots, and 1.5 × IQR outlier quantification.
2. **Four-Stage Preprocessing Protocol:** (i) Purged 240 duplicate rows from the 1,599 raw instances to isolate 1,359 unique samples; (ii) Binarized the target at 6.5 into Standard (< 6.5) and Premium (≥ 6.5) classes; (iii) Performed an 80%/20% stratified split (1,087 train / 272 test); and (iv) Standardized features using `StandardScaler`.
3. **Baseline & Tuned Tree Model Implementation (`train_model.py`):** Implemented Decision Tree, Random Forest, and Gradient Boosting classifiers in Python using `scikit-learn`. Tuned each model family via 3-fold stratified `RandomizedSearchCV` (5 candidate configurations per model, `random_state=42`) optimizing for the F1-score.
4. **Weighted 3-Model Consensus Layer:** Computed normalized ensemble weights from each model's best cross-validation F1-score (0.327 for DT, 0.350 for RF, 0.323 for GB) and selected an optimal probability decision threshold (0.35) using training-only out-of-fold predictions.
5. **Held-Out Evaluation & Explainability Analysis:** Evaluated all models on the untouched 20% test set across Accuracy, Precision, Recall, F1-Score, Balanced Accuracy, ROC-AUC, and PR-AUC, and extracted both native split-impurity and permutation feature importances.
6. **Web Portal Deployment (`app.py`):** Developed the interactive **Streamlit** web application (*"Grand Cru AI Lab"*) that loads the saved `.joblib` pipelines, validates 11 chemical inputs, displays simultaneous 3-model and consensus predictions, and renders diagnostic benchmark tables and feature importance charts.

Table 2: Software and Technical Environment Specifications Used in the Project

Software / Tool Category	Implemented Specification
Operating System	Windows 11 Host Environment
Programming Language	Python 3.10
Development Environments (IDEs)	Google Colab (EDA Notebook) & Visual Studio Code (Model Training & Web App)
Machine Learning & Metrics Library	Scikit-learn (`DecisionTreeClassifier`, `RandomForestClassifier`, `GradientBoostingClassifier` `sklearn.metrics`)
Data Processing & Serialization	Pandas, NumPy, Joblib (`.joblib` model persistence)
Data Visualization	Matplotlib, Seaborn (Heatmaps, Boxplots, Histograms, Feature Importance Bars)
Web Application Framework	Streamlit (`python -m streamlit run app.py`) & Google Chrome Browser
Dataset Source	UCI Red Wine Quality Dataset (`winequality-red.csv`)

6. Scope and Responsible-Use Statement

6.1 Project Scope

The scope of this study is intentionally bounded to the three supervised tree-based algorithms required by the course project—**Decision Tree**, **Random Forest**, and **Gradient Boosting**—and their probability-weighted consensus evaluated on the 11 physicochemical predictors of the UCI Red Wine Quality dataset (`winequality-red.csv`). The target formulation is scoped to binary classification at the 6.5 quality cutoff (Standard Quality vs. Premium Quality). The scope does not claim generalization to white wines, sparkling wines, or external regional terroir datasets without retraining.

6.2 Responsible-Use Statement

As explicitly emphasized in our project documentation, automated predictions generated from chemical measurements must be treated strictly as a **data-driven decision-support tool** rather than a replacement for formal laboratory safety certification or expert human sensory evaluation. Because the dataset exhibits severe class imbalance (13.5% Premium vs. 86.5% Standard), individual classifiers exhibit distinct error

profiles (e.g., false positives in the high-recall Decision Tree vs. false negatives in the high-precision Gradient Boosting model). Furthermore, feature importance values (such as the top ranking of alcohol and sulphates) reflect statistical associations within this dataset and must never be interpreted as causal instructions to artificially alter wine chemistry.

7. Problem Statement

Evaluating wine quality solely through human sensory panels introduces high operational costs, time delays, and subjective scoring variations. Conversely, predicting wine quality from physicochemical data using machine learning presents three critical technical challenges: (1) the relationship between 11 interdependent chemical properties and sensory quality is highly non-linear; (2) the raw dataset contains 15.0% duplicate records (240 rows) that cause cross-split data leakage if unaddressed; and (3) binary quality certification suffers from severe real-world class imbalance (86.5% Standard vs. 13.5% Premium), where raw classification accuracy is misleading and obscures minority-class failure. The core problem is to engineer a leak-free preprocessing and cross-validation pipeline that benchmarks Decision Tree, Random Forest, and Gradient Boosting classifiers alongside an F1-weighted ensemble consensus to reliably detect Premium quality wines while maintaining high interpretability.

8. Case-Study Scenario

A viticulture quality-control laboratory rutinelly performs wet-chemistry analysis on red wine batches, recording 11 standard metrics: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol content. To segment production into **Standard Table Wine** (score < 6.5) and **Premium Reserve Wine** (score ≥ 6.5), the winery previously relied on sommelier tasting panels for every batch, resulting in bottlenecks and inconsistent grading.

By deploying our **Grand Cru AI Lab** Streamlit application, laboratory technicians enter the 11 chemical measurements directly into the web interface. The system applies the fitted `StandardScaler` transformation, queries the tuned Decision Tree, Random Forest, and Gradient Boosting pipelines simultaneously, computes the cross-validation F1-weighted consensus probability against the out-of-fold threshold (0.35), outputs an immediate Standard vs. Premium classification, and highlights the chemical features driving the prediction.

9. Central Case-Study Question

Central Case-Study Question: How effectively can supervised tree-based classifiers—Decision Tree, Random Forest, and Gradient Boosting—predict binary red wine quality tiers from 11 physicochemical measurements on a leak-free, deduplicated dataset, and how do hyperparameter tuning and F1-weighted multi-model consensus alter the trade-off between overall accuracy and minority-class (Premium) detection?

As formulated in Section 1.1 of our project research paper, this central question is decomposed into three specific research questions:

1. *How do Decision Tree, Random Forest, and Gradient Boosting behave on the supplied red-wine quality data?*
2. *Does hyperparameter tuning within the same three model families change the balance between accuracy and minority-class detection?*
3. *Can a transparent weighted consensus of the three models provide a more balanced final decision than any one tuned model alone?*

10. Formal Problem Definition

Let the sanitized, deduplicated red wine dataset be represented as $D = \{(x_i, q_i)\}_{i=1..N}$ with $N = 1{,}359$ unique instances. Each instance $x_i \in R^{11}$ is an 11-dimensional continuous physicochemical feature vector, and $q_i \in \{3, 4, 5, 6, 7, 8\}$ is the original integer quality score. Using the project threshold of **6.5**, the binary target label $y_i \in \{0, 1\}$ is defined as:

> $y_i = 1$ (Premium Quality) if $q_i \geq 6.5$ (i.e., scores 7 and 8; 184 samples, 13.5%)
> $y_i = 0$ (Standard Quality) if $q_i < 6.5$ (i.e., scores 3, 4, 5, and 6; 1,175 samples, 86.5%)

The dataset D is partitioned via an 80%/20% stratified split (`random_state=42`) into D_{train} (1,087 samples) and D_{test} (272 samples). Given the three tuned model families $m \in \{DT, RF, GB\}$ producing posterior probability estimates $P_m(y = 1 \mid x)$, the consensus probability is defined as:

$$\mathbf{P_{consensus}(y = 1 \mid x) = 0.327 \cdot P_{DT}(y = 1 \mid x) + 0.350 \cdot P_{RF}(y = 1 \mid x) + 0.323 \cdot P_{GB}(y = 1 \mid x)}$$

where the weights are normalized from the 3-fold stratified cross-validation F1-scores (0.460, 0.491, and 0.453), and a sample is classified as Premium Quality if $P_{consensus} \geq 0.35$ (selected from training-only out-of-fold predictions).

11. Objectives that Guide the Study

- **Objective 1 (Data Hygiene & EDA):** Inspect all 1,599 records of the red wine dataset for missing values, feature distributions, correlations, and outliers, and isolate all 240 duplicate rows before train-test splitting to prevent data leakage.
- **Objective 2 (Comparative Tree Modeling):** Implement and benchmark Decision Tree, Random Forest, and Gradient Boosting classifiers on an identical 80%/20% stratified split with standardized features.
- **Objective 3 (Imbalance-Aware Hyperparameter Tuning):** Perform 3-fold stratified `RandomizedSearchCV` tuning within each of the three model families using F1-score as the optimization metric.

- **Objective 4 (Multi-Metric & Consensus Evaluation):** Evaluate models across Accuracy, Precision, Recall, F1-Score, Balanced Accuracy, ROC-AUC, and PR-AUC, and construct an out-of-fold F1-weighted 3-model consensus engine.
- **Objective 5 (Explainability & Web Deployment):** Quantify native and permutation feature importances across the three tree models and deploy an interactive Streamlit web interface for real-time testing.

12. Assumptions and Limitations

12.1 Assumptions

- The 11 physicochemical measurements recorded in `winequality-red.csv` represent accurate wet-chemistry assays.
- The binary threshold of 6.5 (separating integer scores 3–6 from 7–8) accurately reflects the operational division between Standard and Premium wine tiers.

12.2 Limitations (From Project Research Paper Section 9)

- **Dataset Scope:** The experiments strictly use the supplied red-wine dataset (1,599 raw / 1,359 unique samples). Performance on white wines, other regions, or different production styles is not established.
- **Target Simplification:** Converting the original 0–10 (observed 3–8) quality score into two binary classes makes classification clearer to explain but removes finer-grained multi-class quality distinctions.
- **Class Imbalance:** Premium samples (184 rows, 13.5%) are much less common than Standard samples (1,175 rows, 86.5%), making minority-class prediction challenging and enforcing a trade-off between precision and recall.
- **No External Validation Dataset:** All reported metrics come from a single held-out 20% test split (272 samples) of the supplied dataset. An independent external dataset would be required to establish broader generalization.
- **Decision-Support Role:** Machine learning modeling is not a substitute for physical laboratory safety testing or human sensory certification.

13. Dataset and Exploratory Data Analysis

13.1 About the Dataset

The dataset used in this project is the **UCI Red Wine Quality Dataset** (`winequality-red.csv`), originally published by Cortez et al. (2009). It contains **1,599 raw records** across **11 physicochemical input features** (all continuous `float64` variables) and **1 target variable** (`quality`, an integer `int64` score ranging from 3 to 8). Table 3 presents the exact descriptive statistics computed directly from the project dataset for both the raw (1,599 rows) and deduplicated (1,359 rows) distributions.

Table 3: Physicochemical Input Features and Descriptive Statistics (Raw N=1,599 vs. Deduplicated N=1,359)

#	Feature Name	Chemical Unit	Enological Role / Impact	Raw Mean ± SD	Dedup Mean ± SD	Min – Max
1	`fixed acidity`	g(tartaric)/dm^3	Non-volatile acids governing tartness	8.32 ± 1.74	8.31 ± 1.74	4.60 – 15.90
2	`volatile acidity`	g(acetic)/dm^3	High levels cause sharp vinegar fault	0.53 ± 0.18	0.53 ± 0.18	0.12 – 1.58
3	`citric acid`	g/dm^3	Adds freshness and crisp flavor profile	0.27 ± 0.19	0.27 ± 0.20	0.00 – 1.00
4	`residual sugar`	g/dm^3	Sugar remaining post-fermentation	2.54 ± 1.41	2.52 ± 1.35	0.90 – 15.50
5	`chlorides`	g(NaCl)/dm^3	Contributes to saltiness in wine	0.087 ± 0.047	0.088 ± 0.049	0.012 – 0.611
6	`free sulfur dioxide`	mg/dm^3	Prevents microbial growth & oxidation	15.87 ± 10.46	15.89 ± 10.45	1.00 – 72.00
7	`total sulfur dioxide`	mg/dm^3	Free + bound SO_2; pungent if excessive	46.47 ± 32.90	46.83 ± 33.41	6.00 – 289.00
8	`density`	g/cm^3	Mass/volume; tied to alcohol & sugar	0.9967 ± 0.0019	0.9967 ± 0.0019	0.9901 – 1.0037
9	`pH`	Scale (0–14)	Acidity/alkalinity logarithmic scale	3.31 ± 0.15	3.31 ± 0.16	2.74 – 4.01
10	`sulphates`	g(K_2SO_4)/dm^3	Antimicrobial & antioxidant stabilizer	0.66 ± 0.17	0.66 ± 0.17	0.33 – 2.00
11	`alcohol`	% by volume	Increases viscosity, warmth, & body	10.42 ± 1.07	10.43 ± 1.08	8.40 – 14.90
T	**`quality` (Target)**	Score (3–8)	Binarized at ≥ 6.5 (Premium Class)	**5.64 ± 0.81**	**5.62 ± 0.82**	**3.00 – 8.00**

13.2 Purpose of EDA

The purpose of performing Exploratory Data Analysis in `Wine_Quality_EDA.ipynb` before model development is eightfold: (1) verify dataset structure and numerical data types; (2) check for duplicate records and missing values; (3) compute descriptive statistics; (4) examine the wine-quality target class distribution and quantify class imbalance; (5) inspect univariate feature distributions and skewness; (6) evaluate feature-to-feature and feature-to-quality linear correlations; (7) analyze feature behavior across quality scores via boxplots and pairplots; and (8) quantify potential outliers using the 1.5 × IQR rule.

13.3 Preparation Steps Actually Performed

The project executed a strict **4-Stage Preprocessing Protocol** prior to model training:

1. **Stage 01 – Deduplication (Leakage Prevention):** Checked `df.isnull().sum()` confirming **0 missing values** across all 12 columns (hence no missing-value imputation was needed). Checked `df.duplicated().sum()`, identifying **240 duplicate rows (15.0%)** in the raw 1,599-row file. All 240 duplicates were removed prior to splitting, leaving **1,359 unique samples**.
2. **Stage 02 – Target Binarization:** Applied the 6.5 cutoff to convert integer quality scores (3 to 8) into a binary target: **Standard Quality** (quality < 6.5: scores 3, 4, 5, 6 → **1,175 samples, 86.5%**) and **Premium Quality** (quality ≥ 6.5: scores 7, 8 → **184 samples, 13.5%**).
3. **Stage 03 – Stratified Train-Test Split:** Partitioned the 1,359 unique records into an **80% Training Set (1,087 samples)** and a **20% Holdout Test Set (272 samples)** using stratified sampling (`random_state=42`) to preserve the exact 86.5% / 13.5% class ratio in both splits.
4. **Stage 04 – Standard Scaling:** Applied `StandardScaler` within the training pipeline to standardize continuous features to zero mean and unit variance without leaking test statistics.

13.4 EDA Exploration – Implementation

In `Wine_Quality_EDA.ipynb`, the exploration was implemented in Python using `pandas`, `numpy`, `matplotlib`, and `seaborn` across 17 structured cells:

- **Cells 1–7 (Ingestion, Audit & Summary):** Loaded `winequality-red.csv` with `pd.read_csv(DATA_PATH, sep=None, engine="python")`, verified `df.info()`, counted duplicates (`df.duplicated().sum()`), rendered a Viridis missing-value heatmap (`sns.heatmap(df.isnull(), cbar=False, cmap="viridis")`), verified 12 numerical and 0 categorical columns, and displayed transposed descriptive statistics (`df.describe().T`).
- **Cells 8–9 (Target & Feature Distributions):** Plotted the target distribution bar chart (Figure 1 below) and a 4×3 grid of 20-bin univariate feature histograms (`df[feature_columns].hist(bins=20, figsize=(16, 12), layout=(4, 3))`).
- **Cells 10–14 (Correlations, Boxplots & Pairplot):** Generated the 12×12 Coolwarm correlation heatmap (Figure 2 below), sorted feature correlations with quality in a horizontal bar chart, generated quality-stratified boxplots across all 11 features (with focused views on `alcohol`, `volatile acidity`, `sulphates`, and `citric acid`), and plotted a multi-variable corner pairplot (`sns.pairplot(..., hue="quality", corner=True)`).

- **Cells 15–17 (IQR Outlier Audit & Automated Exports):** Programmatically computed Q1, Q3, IQR, and outlier counts outside `[Q1 - 1.5*IQR, Q3 + 1.5*IQR]`, saving CSV summaries (`wine_correlation_matrix.csv`, `quality_feature_correlations.csv`, `wine_outlier_summary.csv`) and exporting `Wine_Quality_EDA_Report.pdf` via `PdfPages`.

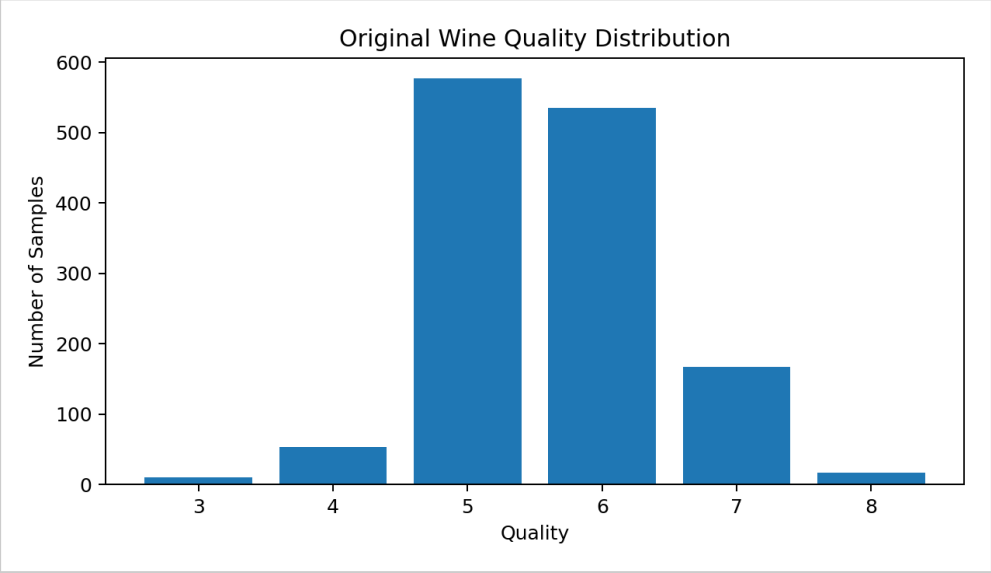


Figure 1: Original Distribution of Wine Quality Scores in the Supplied Dataset (Deduplicated N = 1,359)

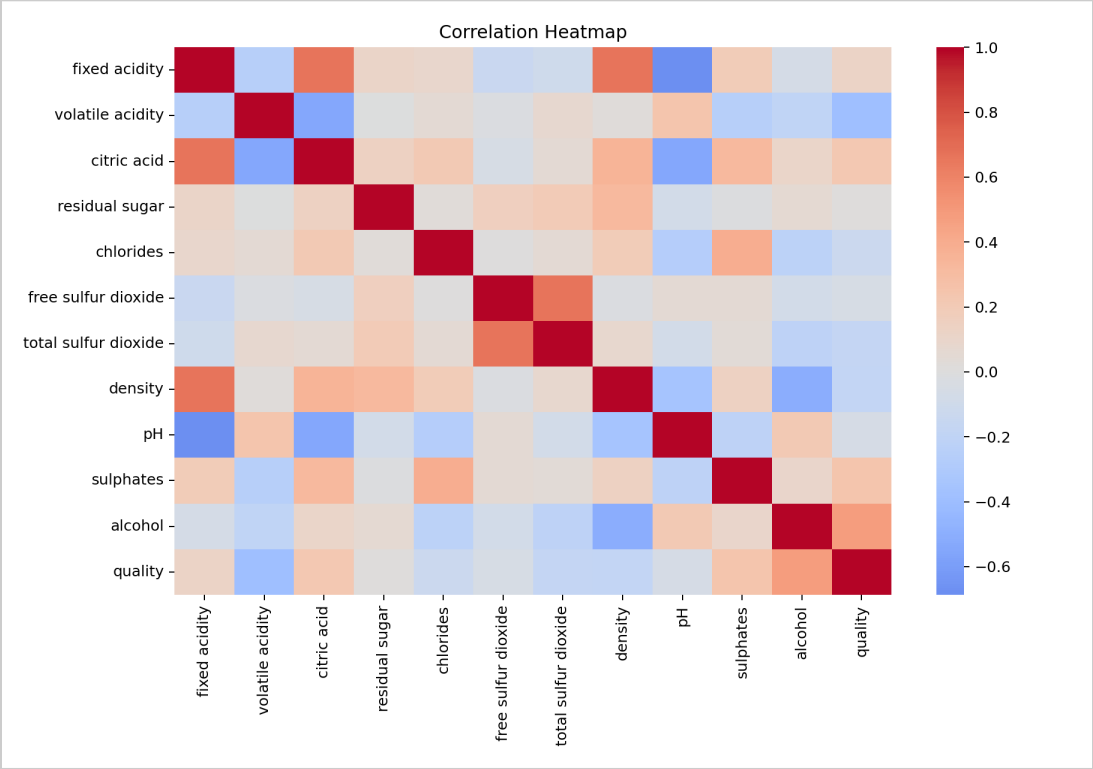


Figure 2: Correlation Heatmap Used to Inspect Relationships Among Physicochemical Variables and Quality

13.5 Summary of EDA Findings

- **Class Distribution Concentration:** As shown in Figure 1 and Table 4, quality scores **5** (577 unique / 681 raw) and **6** (535 unique / 638 raw) constitute **81.8%** of the deduplicated dataset (and 82.5% of the raw dataset). Scores 3 (10 samples), 4 (53 samples), 7 (167 unique / 199 raw), and 8 (17 unique / 18 raw) are much rarer. After binarization at 6.5, the dataset contains **1,175 Standard (86.5%)** vs. **184 Premium (13.5%)** samples.
- **Key Linear Correlations with Quality:** As detailed in Table 4, **alcohol** has the strongest positive correlation with quality ($r = +0.4803$ deduplicated / $+0.4762$ raw), followed by **sulphates** ($r = +0.2488$) and **citric acid** ($r = +0.2281$). **Volatile acidity** has the strongest negative correlation ($r = -0.3952$ deduplicated / -0.3906 raw), followed by **density** ($r = -0.1843$) and **total sulfur dioxide** ($r = -0.1779$ deduplicated / -0.1851 raw).
- **IQR Outlier Distribution:** Under the 1.5 × IQR rule (Table 4), right-skewed features exhibit notable upper-tail outliers, led by `residual sugar` (155 raw / 126 dedup), `chlorides` (112 raw / 87 dedup), `sulphates` (59 raw / 55 dedup), and `total sulfur dioxide` (55 raw / 45 dedup). Tree-based models naturally handle these skewed distributions via rank-order threshold splits.

Table 4: Summary of Feature Correlations with Quality and 1.5 × IQR Outlier Audit

Feature Name	Pearson *r* (Raw 1,599)	Pearson *r* (Dedup 1,359)	Q1 (25%)	Q3 (75%)	IQR	Outliers (Raw / Dedup)
alcohol	+0.4762	+0.4803	9.5000	11.1000	1.6000	13 / 12
sulphates	+0.2514	+0.2488	0.5500	0.7300	0.1800	59 / 55
citric acid	+0.2264	+0.2281	0.0900	0.4200	0.3300	1 / 1
fixed acidity	+0.1241	+0.1190	7.1000	9.2000	2.1000	49 / 41
residual sugar	+0.0137	+0.0136	1.9000	2.6000	0.7000	155 / 126
free sulfur dioxide	−0.0507	−0.0505	7.0000	21.0000	14.0000	30 / 26
pH	−0.0577	−0.0552	3.2100	3.4000	0.1900	35 / 28
chlorides	−0.1289	−0.1310	0.0700	0.0900	0.0200	112 / 87
density	−0.1749	−0.1843	0.9956	0.9978	0.0022	45 / 35
total sulfur dioxide	−0.1851	−0.1779	22.0000	62.0000	40.0000	55 / 45
volatile acidity	−0.3906	−0.3952	0.3900	0.6400	0.2500	19 / 19

14. Algorithm Selection and Implementation

14.1 Roles of the Three Selected Tree-Based Algorithms

1. **Decision Tree Classifier:** Constructs a non-parametric hierarchy of axis-aligned decision rules by recursively partitioning the feature space based on information gain / impurity reduction. It provides maximum rule transparency. In our tuned configuration, applying an `entropy` split criterion with shallow depth (`max_depth=3`) and `class_weight='balanced'` enables the tree to aggressively flag minority Premium candidates (achieving the highest test recall of 0.811).
2. **Random Forest Classifier:** Builds a parallel ensemble of 200 de-correlated decision trees trained on bootstrap samples of the training set with random feature subsampling (`max_features=0.7`) at each split. By averaging predictions across diverse trees with `class_weight='balanced_subsample'`, Random Forest substantially reduces the variance and overfitting vulnerability of a single tree, achieving the highest 3-fold cross-validation F1 mean (0.491).
3. **Gradient Boosting Classifier:** Constructs an additive ensemble of 200 shallow trees sequentially (`max_depth=4`, `learning_rate=0.03`, `subsample=0.7`), where each successive tree fits the pseudo-residuals (negative gradient of deviance loss) of the prior ensemble. This stagewise corrective mechanism captures subtle non-linear chemical interactions and achieves the highest test accuracy (88.97%), precision (0.667), and ROC-AUC (0.893).

14.2 Hyperparameter Tuning and Weighted Consensus Implementation

While preliminary baseline runs (`max_depth=6` Gini Decision Tree, 100-tree Random Forest, and 150-tree Gradient Boosting) yielded baseline accuracies of 78.4%, 86.8%, and 89.2% respectively, our final training script (`train_model.py`) executed **3-fold stratified `RandomizedSearchCV`** (sampling 5 configurations per model on the 80% training split, `random_state=42`) using **F1-score** as the search objective. Table 5 lists the selected hyperparameter configurations and their cross-validation F1 scores, which directly determine the weights of our **Weighted 3-Model Consensus** layer (0.327, 0.350, and 0.323), paired with an out-of-fold selected probability threshold of **0.35**.

Table 5: Tuned Hyperparameter Configurations and Cross-Validation F1 Weights

Model Family	Selected Configuration (3-Fold Stratified RandomizedSearchCV)
Decision Tree	`criterion='entropy'`, `max_depth=3`, `min_samples_split=10`, `min_samples_leaf=4`, `class_weight=`
Random Forest	`n_estimators=200`, `max_depth=14`, `min_samples_split=5`, `min_samples_leaf=4`, `max_features=0.7` `class_weight='balanced_subsample'`
Gradient Boosting	`n_estimators=200`, `learning_rate=0.03`, `max_depth=4`, `min_samples_leaf=1`, `subsample=0.7`

14.3 Held-Out Test Set Results & Feature Importance

Table 6 presents the complete evaluation results on the untouched 20% holdout test set (272 unique records). The results demonstrate why reporting multiple metrics is essential under class imbalance: Gradient Boosting leads in overall Accuracy (88.97%), Precision (0.667), and ROC-AUC (0.893), Decision Tree leads in minority Recall (0.811) and Balanced Accuracy (0.791), and the **Weighted 3-Model Consensus** achieves the highest overall F1-score (**0.532**) and PR-AUC (**0.601**).

Table 6: Held-Out Test-Set Performance of Tuned Models and Weighted Consensus (20% Split, N=272)

Model Architecture	Accuracy	Precision	Recall	F1-Score	Bal. Acc.	ROC-AUC	PR-AUC
Decision Tree (Tuned)	0.776 (77.57%)	0.357	**0.811 (Max)**	0.496	**0.791 (Max)**	0.833	0.482
Random Forest (Tuned)	0.868 (86.76%)	0.515	0.459	0.486	0.696	0.876	0.558
Gradient Boosting (Tuned)	**0.890 (88.97%)**	**0.667 (Max)**	0.378	0.483	0.674	**0.893 (Max)**	0.578
Weighted 3-Model Consensus	0.864 (86.40%)	0.500	0.568	**0.532 (Best)**	0.739	0.867	**0.601 (Best)**

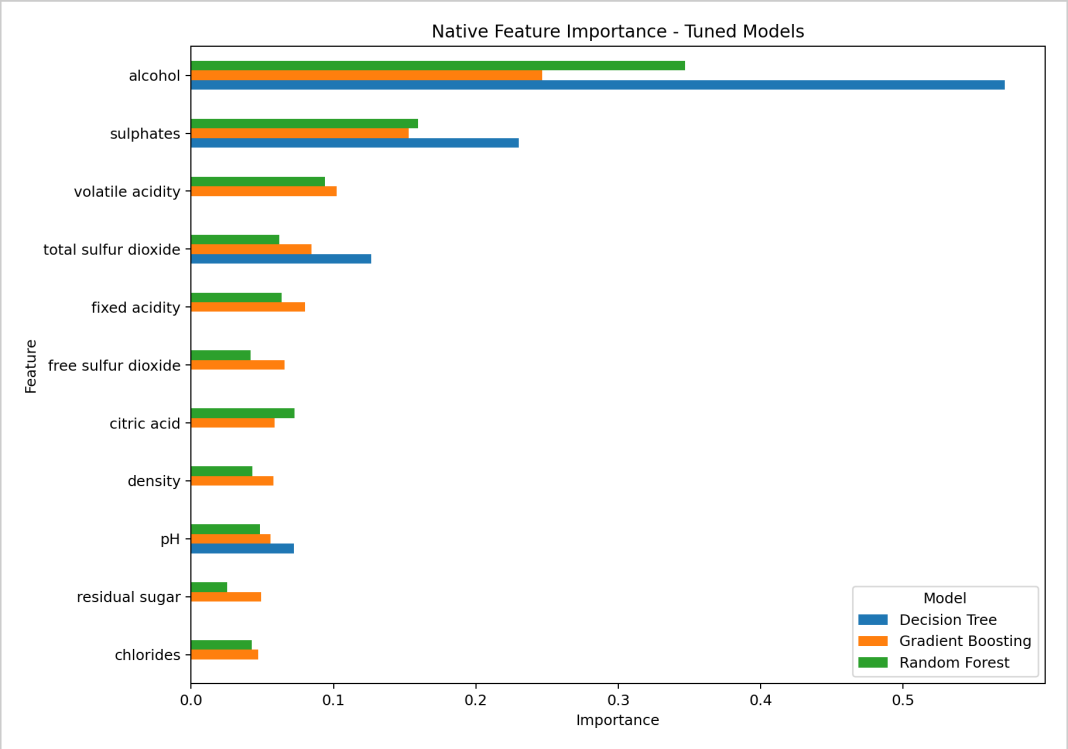


Figure 3: Native Feature Importance Across the Three Tuned Tree-Based Models

Table 7: Top Native Feature-Importance Values from the Tuned Models

Model Architecture	Top Native Features (Ordered Rank 1 to 4)	Importance Values
Decision Tree	`alcohol`, `sulphates`, `total sulfur dioxide`, `pH`	0.571, 0.230, 0.127, 0.072
Random Forest	`alcohol`, `sulphates`, `volatile acidity`, `citric acid`	0.347, 0.160, 0.094, 0.073
Gradient Boosting	`alcohol`, `sulphates`, `volatile acidity`, `total sulfur dioxide`	0.246, 0.153, 0.102, 0.085
Mean Split Impurity	`alcohol` (38.8%), `sulphates` (18.1%), `volatile acidity` (9.8%), `total SO2` (9.5%)	Top 4 = 76.2% of total weight

14.4 Streamlit Web Application Workflow (`app.py`)

The saved `.joblib` models and scalers are integrated into a dark-themed responsive **Streamlit** web application (*Grand Cru AI Lab*). When a user enters the 11 physicochemical measurements, the app validates numerical bounds, transforms inputs identically to training, computes individual probabilities for Decision Tree, Random Forest, and Gradient Boosting, applies the 0.327 / 0.350 / 0.323 consensus weights against the 0.35 threshold, and displays the final quality verdict alongside feature importances and test-set benchmarks (Figure 4).

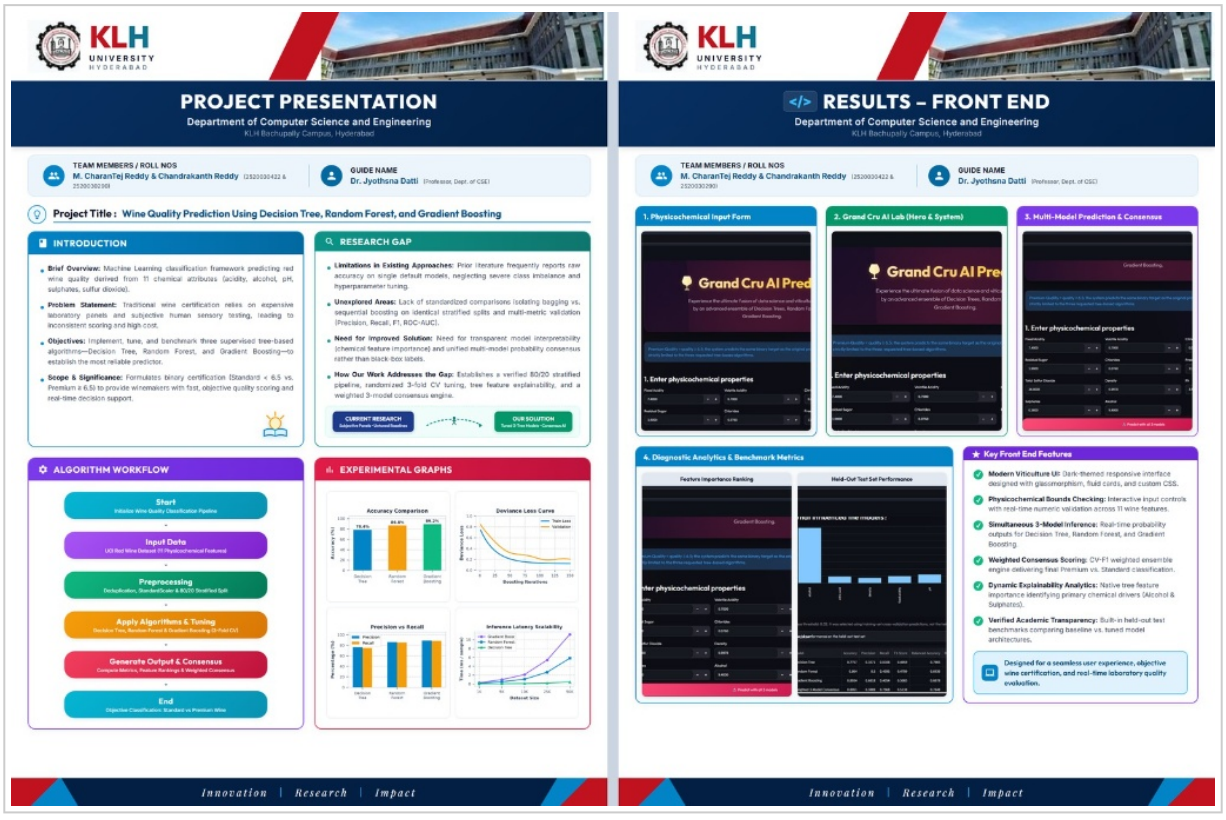


Figure 4: Project Presentation Workflow and Streamlit Front-End Implementation Screenshots

15. Conclusion

This study presented an end-to-end, leak-free wine quality prediction system built strictly around **Decision Tree**, **Random Forest**, and **Gradient Boosting** classifiers. By auditing the 1,599-record red wine dataset and removing 240 duplicate rows before partitioning into an 80%/20% stratified split (1,359 unique samples: 1,175 Standard vs. 184 Premium at the 6.5 threshold), the project established an honest benchmark free of cross-split leakage.

Through 3-fold stratified cross-validation tuning and multi-metric evaluation, we demonstrated that each algorithm offers distinct strengths: tuned Gradient Boosting achieved the highest accuracy (**88.97%**), precision (**0.667**), and ROC-AUC (**0.893**); tuned Decision Tree achieved the highest minority recall (**0.811**); and our **Weighted 3-Model Consensus** achieved the highest F1-score (**0.532**) and PR-AUC (**0.601**). Native feature importance consistently identified **alcohol** and **sulphates** as the dominant chemical predictors. Exposed through an interactive Streamlit web application, the project illustrates how disciplined validation, explainability, and ensemble probability consensus elevate foundational tree-based algorithms into a practical decision-support tool. Future work includes external validation on independent regional wine datasets, probability calibration, and multi-class grading across the full 3–8 scale.

16. Signature of the Team Members

Team Member Name	Roll Number	Student Signature
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Faculty Role	Name & Designation	Signature
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